Annex 3. Côte d'Ivoire: Joint Bank-Fund Debt Sustainability Analysis

The debt sustainability analysis (DSA) shows that Côte d'Ivoire is at a high risk of debt distress. [15] *Under the external DSA baseline scenario, NPV-based indicators are projected to stay above the relevant indicative thresholds over the next few years. Alternative scenarios illustrate the impact of additional HIPC, MDRI and beyond-HIPC debt relief at the completion point. Under these scenarios, debt burden indicators would improve significantly. The inclusion of domestic debt does not alter the assessment of Côte d'Ivoire's risk of debt distress.*

1. **This DSA is based on the Debt Sustainability Framework (DSF) for Low-Income Countries.** The DSA presents the projected path of Côte d'Ivoire's external and public sector debt burden indicators, and assesses the country's risk of external debt distress. Methodologically, the DSA differs from the Debt Relief Analysis (DRA) in the decision point document in that it compares the evolution over the projection period of debt burden indicators against policy-dependent indicative thresholds. In contrast, under the DRA, the historical debt burden indicators are compared to uniform thresholds in order to calculate the amount of HIPC debt relief that Côte d'Ivoire qualifies for in the context of the HIPC Initiative.[16]

2. **The last DSA for Côte d'Ivoire was prepared in 2007 and assessed Côte d'Ivoire as being in debt distress**.[17] At that time, Côte d'Ivoire was accumulating arrears to a number of creditors and several debt burden indicators showed substantial breaches of their respective indicative thresholds. Compared to 2007 DSA results, the analysis below finds improved debt dynamics as a consequence of projected improvements in the macroeconomic policy framework, large arrears clearance operations, and HIPC Initiative interim debt relief.

C. Baseline Assumptions

3. **The assumptions of the DSA are consistent with projections of the HIPC Initiative DRA and with the PRGF arrangement** (Box 1). After several years of below-trend growth, real GDP growth is expected to have improved to 2.3 percent in 2008

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[15] This DSA has been prepared jointly by the World Bank and Fund staffs using the Debt Sustainability Framework (DSF) for Low Income Countries (see "Applying the Debt Sustainability Framework for Low-Income Countries Post Debt Relief" (www.imf.org, 11/06/06 and IDA/SecM2006-0564, 8/11/06). Côte d'Ivoire's quality of policies and institutions, as measured by the average World Bank's Country Policy and Institutional Assessment (CPIA) for the period 2005-2007 (2.5), places it as a "weak performer." The corresponding indicative thresholds for the external debt indicators are 30 percent for the NPV of debt-to-GDP ratio, 100 percent for the NPV of debt-to-exports ratio, 200 percent for NPV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio and 25 percent for the debt service-to-revenue ratio.

[16] In addition, the results of the LIC DSA differ from the HIPC DRA because of methodological differences related to the definition of the discount rates and the exchange rates.

[17] Côte d'Ivoire: 2007 Article IV Consultation and Request for Emergency Post-Conflict Assistance - Staff Report; Public Information Notice on the Executive Board Discussion; and Statement by the Executive Director for Côte d'Ivoire.

and increase further to 3.7 percent in 2009. Growth continues to be driven by construction, crude oil, food production, and telecommunications. Overall agricultural exports stagnated, as cashew nuts expanded while coffee and cotton declined because of structural problems. The projected pick-up in growth to 4.7 percent over the medium term reflects full political normalization, including successful elections, restoration of investor confidence, return to full capacity utilization and normal levels of public and private investment, and structural reforms. These projections are subject to downside risks related to the global environment and commodity prices.

4. **After some acceleration, inflation is expected to return to the WAEMU target. CPI** inflation accelerated to 6.3 percent on average in 2008, fueled by import food prices and domestic price adjustments. The surge in retail food prices was tempered by the reduction in VAT and import duties on essential food items, effective since April. With declines in food and oil prices, inflation is projected to moderate in 2009 and be contained over the medium term, anchored in the WAEMU currency union and fiscal prudence.

5. **Macroeconomic stability will be anchored on a prudent fiscal policy and structural reforms.** The fiscal strategy aims at continued fiscal consolidation and addressing social and reconstruction needs. The 2009 budget aims at creating additional fiscal space for growth-enhancing investment and pro-poor social expenditure. The authorities intend to rebalance the composition of spending by allowing higher social expenditure while reducing sovereign spending and non-essential investment, including on the large-scale investment projects related to the transfer of the capital to Yamoussoukro. The authorities also plan to enhance transparency and governance, achieve debt sustainability and improve overall public finance management.

7. **The baseline scenario reflects the full delivery of traditional debt relief as well as interim HIPC assistance.**[18] Consistent with the DSF guidelines, the baseline does not reflect the delivery of HIPC, MDRI and bilateral or multilateral beyond-HIPC assistance after the completion point.[19] The evolution of Côte d'Ivoire's debt indicators after the full impact of debt relief under the HIPC Initiative and MDRI is presented as an alternative scenario (Figure 3). Furthermore, given Côte d'Ivoire's extensive level of debt to the Paris Club, an additional scenario presenting the impact of HIPC, MDRI and debt relief beyond HIPC by Paris Club creditors is also presented.

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[18] The authorities cleared arrears to the World Bank Group in 2008 and AfDB in early 2009 and agreed on modalities for arrears clearances with EIB/EU.

[19] See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (www.imf.org and IDA/SECM2007/0226, 03/05/2007).

D. Debt Sustainability Analysis

External Debt Sustainability

8. **Under the baseline scenario, Côte d'Ivoire's external debt indicators remain above their relevant indicative thresholds** (Table 1a, Figure 1). The NPV of public and publicly guaranteed (PPG) debt declines from 74 percent of GDP in 2007 to 63 percent of GDP in 2008 and stays above the 30 percent indicative threshold until 2013. The decline in 2008 is primarily on the back of robust GDP growth. The NPV of external debt-to-revenue ratio moves below the 200 percent indicative threshold in 2013. Conversely, debt service ratios generally remain below their respective indicative thresholds over the whole projection period, reflecting the high degree of concessionality of Côte d'Ivoire's debt. However, the debt service-to-revenue ratio breaches its indicative threshold (25 percent of GDP) in 2012–14, reflecting the expiration of the grace period assumed for the restructured debt owed to the London Club and the Paris Club.

9. **Côte d'Ivoire's external debt outlook is subject to considerable vulnerabilities** (Table 1b, Figure 1). Given the impact of distortions on trend caused by the civil conflict in Côte d'Ivoire, the sensitivity analysis is based on regional averages and standard deviations for all relevant indicators. The sensitivity analysis highlights Côte d'Ivoire's vulnerability to a one-time 30 percent nominal depreciation of the CFA in 2009, a reduction in official and private transfers and net FDI, and to new public external borrowing on less favorable terms.

10. **Alternative scenarios assume additional delivery of debt relief at the completion point** (Figure 3). Debt relief under the HIPC Initiative, MDRI and possible bilateral and multilateral beyond-HIPC assistance, would significantly improve Côte d'Ivoire's external debt outlook. Reaching the completion point, which is assumed in 2011, and the resulting irrevocable debt relief would reduce all external debt indicators below the relevant indicative thresholds.

Public Sector Debt Sustainability

11. **If domestic public debt is included in the analysis, Côte d'Ivoire's debt situation deteriorates modestly** (Table 2a, Figure 2)**.** However, public debt ratios would fall over the long run owing to the projected improvement in the macroeconomic outlook and high concessionality of new borrowing under the baseline scenario.

12. **Public debt dynamics are vulnerable to shocks** (Table 2b, Figure 2). Public debt indicators are sensitive to the assumptions on GDP growth, and a one-time depreciation of the CFA.

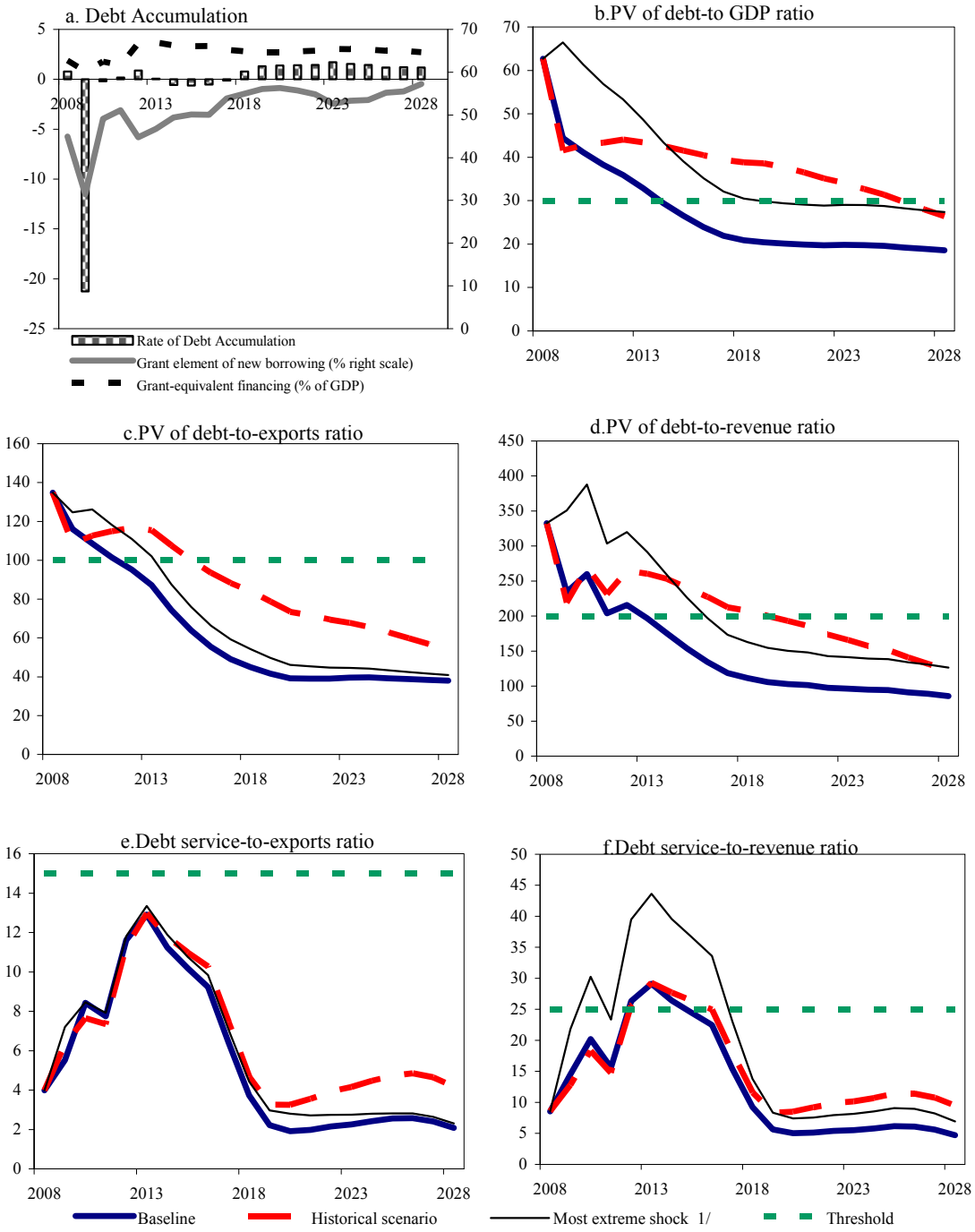
E. Conclusion

13. **In the staff's view, Côte d'Ivoire is at high risk of debt distress.** This compares to the "in debt distress", which was the outcome of the 2007 DSA. The change is primarily on account of multilateral arrears clearance and assumptions on interim HIPC debt relief. In the

current LIC DSA, the PV of external debt-to-GDP and the PV of debt-to-revenue ratios, although declining, remain above the indicative thresholds for some time. Alternative scenarios and bound tests emphasize the vulnerability of Côte d'Ivoire's external debt outlook. The inclusion of domestic debt does not alter the assessment of Côte d'Ivoire's risk of debt distress.

14. **A sustainable external debt position can be achieved based on sound macroeconomic policies, debt relief and a policy of concessional borrowing.** Sustained increased growth, solid export and fiscal revenue performance and prudent debt management are important for achieving debt sustainability. Debt relief under the HIPC Initiative, MDRI and beyond-HIPC assistance significantly improves Côte d'Ivoire's external debt outlook.

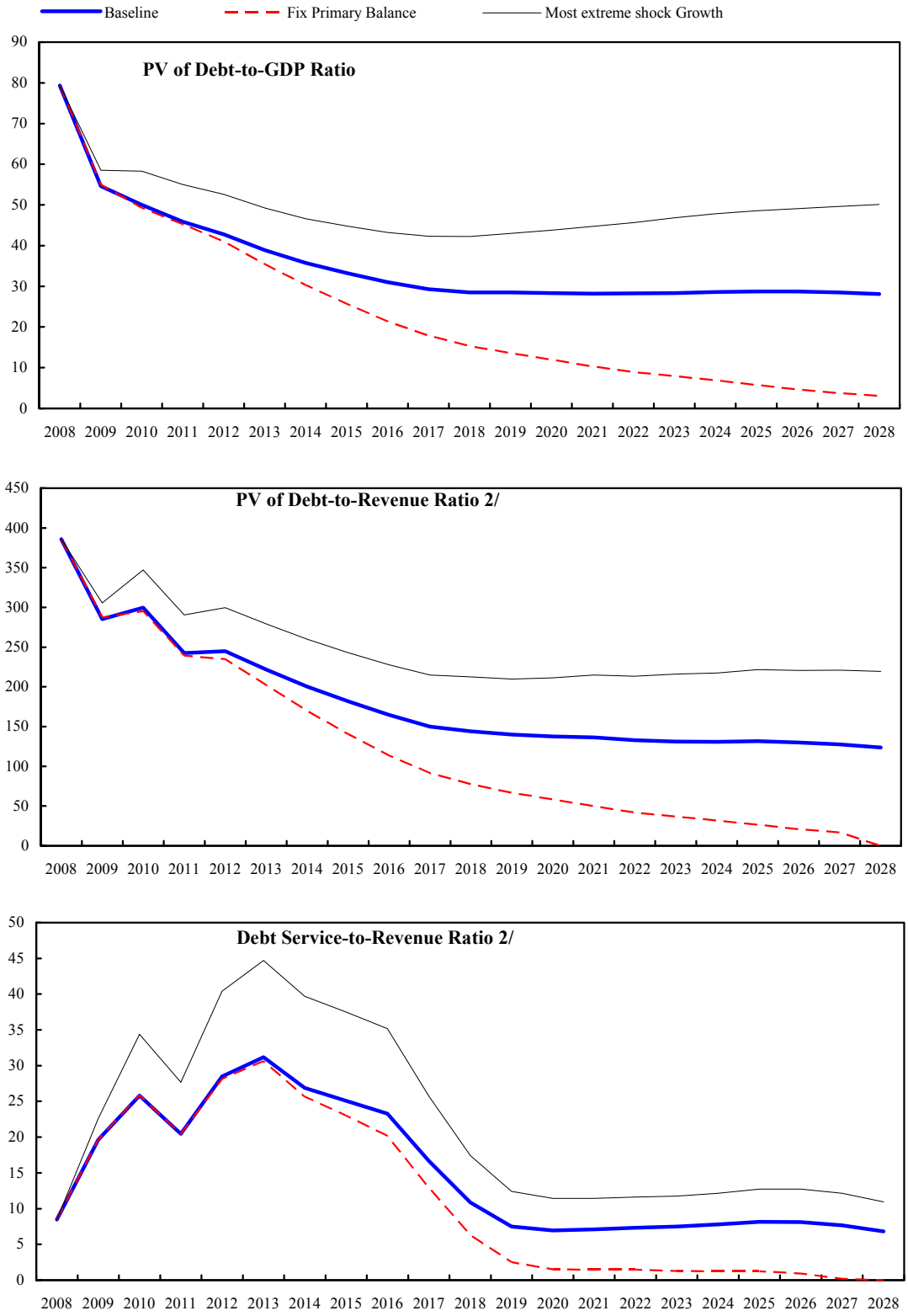
Figure 1. Cote d'Ivoire: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2008-2028 1/



Source: Staff projections and simulations.

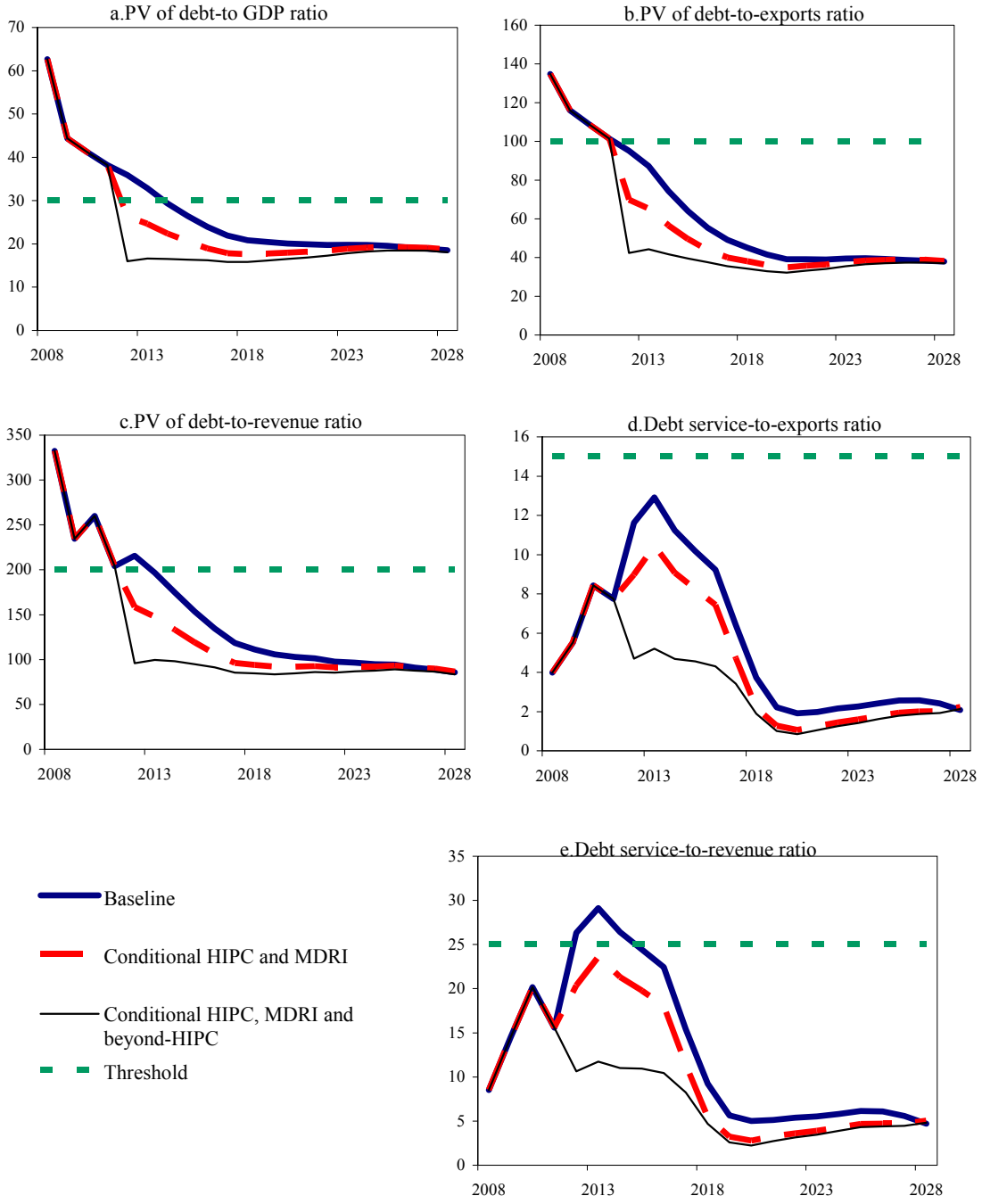
1/ The most extreme stress test is the test that yields the highest ratio in 2018. In figure b. it corresponds to a One-time depreciation shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in picture f. to a One-time depreciation shock

Figure 2.Cote d'Ivoire: Indicators of Public Debt Under Alternative Scenarios, 2008-2028 1/



Sources: Country authorities; and Fund staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2018.
2/ Revenues are defined inclusive of grants.

Figure 3. Cote d'Ivoire: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2008-2028 1/



Source: Staff projections and simulations.

Table 1a.: External Debt Sustainability Framework, Baseline Scenario, 2005-2028 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008-2013 Average	2018	2028	2014-2028 Average
External debt (nominal) 1/	**44.8**	**38.4**	**100.8**			**92.4**	**68.8**	**63.4**	**58.9**	**56.8**	**54.1**		**40.3**	**36.0**	
o/w public and publicly guaranteed (PPG)	0.0	0.0	67.8			65.2	40.6	38.0	36.1	36.5	36.0		30.1	32.1	
Change in external debt	-6.0	-6.4	62.4			-8.4	-23.6	-5.4	-4.5	-2.1	-2.7		-1.6	-0.7	
Identified net debt-creating flows	-4.2	-6.5	-5.7			-6.4	-5.9	-3.2	-1.6	-0.3	0.6		2.6	4.3	
Non-interest current account deficit	**-4.0**	**-6.0**	**-2.1**	**-5.0**	**3.6**	**-4.0**	**-2.8**	**-1.2**	**0.8**	**2.3**	**3.6**		**5.3**	**6.6**	5.8
Deficit in balance of goods and services	-7.5	-10.3	-5.9			-7.7	-6.8	-4.5	-2.5	-0.7	0.8		2.1	4.0	
Exports	51.1	52.7	47.8			46.5	38.2	37.9	37.7	37.8	37.7		46.3	48.9	
Imports	43.6	42.4	41.9			38.8	31.4	33.4	35.2	37.1	38.5		48.5	52.9	
Net current transfers (negative = inflow)	2.8	3.1	2.1	3.1	0.5	1.1	2.3	2.1	2.0	1.8	1.4		1.4	1.4	1.4
o/w official	0.1	0.2	-0.8			-1.4	-0.5	-0.7	-0.8	-1.0	-1.0		-1.0	-1.0	
Other current account flows (negative = net inflow)	0.7	1.2	1.7			2.6	1.8	1.2	1.3	1.1	1.3		1.7	1.1	
Net FDI (negative = inflow)	**-1.9**	**-1.8**	**-2.2**	**-2.1**	**0.5**	**-2.1**	**-2.1**	**-2.2**	**-2.3**	**-2.4**	**-2.3**		**-1.8**	**-1.1**	-1.5
Endogenous debt dynamics 2/	**1.7**	**1.3**	**-1.4**			**-0.4**	**-1.1**	**0.3**	**0.0**	**-0.1**	**-0.6**		**-1.0**	**-1.2**	
Contribution from nominal interest rate	4.5	3.9	3.3			1.6	2.6	3.0	2.7	2.7	2.4		1.1	0.5	
Contribution from real GDP growth	-0.9	-0.3	-0.5			-2.0	-3.7	-2.7	-2.7	-2.8	-3.0		-2.1	-1.7	
Contribution from price and exchange rate changes	-1.8	-2.2	-4.2			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-1.9**	**0.1**	**68.1**			**-2.0**	**-17.7**	**-2.2**	**-3.0**	**-1.8**	**-3.3**		**-4.2**	**-5.1**	
o/w exceptional financing	-4.5	-4.0	-3.0			0.1	-21.7	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	106.6			89.9	72.6	66.6	61.1	56.2	51.0		31.0	22.5	
In percent of exports	...	...	222.9			193.2	189.8	175.7	162.0	149.0	135.3		67.0	45.9	
PV of PPG external debt	**...**	**...**	**73.5**			**62.7**	**44.4**	**41.1**	**38.2**	**35.9**	**32.9**		**20.9**	**18.6**	
In percent of exports	**...**	**...**	**153.8**			**134.8**	**116.1**	**108.6**	**101.4**	**95.1**	**87.2**		**45.0**	**37.9**	
In percent of government revenues	**...**	**...**	**383.9**			**332.6**	**234.4**	**259.9**	**204.0**	**215.7**	**196.8**		**111.3**	**85.8**	
Debt service-to-exports ratio (in percent)	**13.8**	**13.9**	**17.6**			**14.5**	**16.2**	**20.5**	**18.2**	**21.7**	**20.1**		**7.0**	**3.7**	
PPG debt service-to-exports ratio (in percent)	**4.1**	**4.2**	**8.5**			**4.0**	**5.5**	**8.4**	**7.7**	**11.6**	**12.9**		**3.7**	**2.1**	
PPG debt service-to-revenue ratio (in percent)	**12.3**	**11.9**	**21.1**			**8.5**	**14.5**	**20.2**	**15.6**	**26.3**	**29.1**		**9.2**	**4.7**	
Total gross financing need (Billions of U.S. dollars)	1.5	1.1	1.9			1.0	1.8	2.0	2.4	3.2	3.6		4.0	8.2	
Non-interest current account deficit that stabilizes debt ratio	2.0	0.4	-64.5			4.4	20.8	4.2	5.3	4.4	6.2		6.9	7.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.9	0.7	1.6	0.4	2.5	2.3	3.7	4.2	4.7	5.3	5.8	4.3	5.4	4.9	5.1
GDP deflator in US dollar terms (change in percent)	3.8	5.3	12.3	5.3	9.5	15.9	-10.0	3.1	3.3	3.4	3.5	3.2	3.4	3.1	3.2
Effective interest rate (percent) 5/	9.3	9.2	9.8	8.0	1.9	1.9	2.6	4.7	4.6	5.1	4.6	3.9	2.9	1.5	2.6
Growth of exports of G&S (US dollar terms, in percent)	11.2	9.4	3.6	7.5	12.4	15.3	-23.3	6.5	7.6	9.0	9.3	4.1	13.1	7.4	10.4
Growth of imports of G&S (US dollar terms, in percent)	17.1	3.1	12.9	8.3	12.8	9.7	-24.5	14.3	13.9	14.8	13.7	7.0	14.7	6.8	10.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	45.0	31.5	49.1	51.1	44.8	46.8	44.7	55.0	57.3	53.9
Government revenues (excluding grants, in percent of GDP)	17.0	18.4	19.2			18.9	18.9	15.8	18.7	16.7	16.7		18.7	21.6	19.6
Aid flows (in Billions of US dollars) 7/	0.2	0.1	0.1			0.5	0.2	0.5	0.6	1.9	2.1		2.0	4.1	
o/w Grants	0.2	0.1	0.1			0.4	0.0	0.2	0.0	0.2	0.3		0.5	1.1	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.2	0.3	0.5	1.7	1.9		1.5	3.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.9	0.8	1.8	1.4	3.6	3.7		2.8	2.7	3.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			89.3	37.8	65.2	54.3	51.0	53.1		65.9	68.6	64.4
Memorandum items:															
Nominal GDP (Billions of US dollars)	16.4	17.4	19.8			23.5	21.9	23.6	25.5	27.7	30.4		46.5	102.5	
Nominal dollar GDP growth	5.7	6.0	14.0			18.6	-6.7	7.4	8.1	8.9	9.5	7.6	8.9	8.1	8.4
PV of PPG external debt (in Billions of US dollars)			14.6			14.7	9.7	9.7	9.7	10.0	10.0		9.7	19.0	
(PVt-PVt-1)/GDPt-1 (in percent)						0.8	-21.3	-0.2	0.2	0.9	0.1	-3.3	0.8	1.2	0.9

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Cote d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028
(In percent)

	Projections												
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	**2018**	2023	2028
PV of debt-to GDP ratio													
Baseline	63	44	41	38	36	33	29	27	24	22	**21**	20	19
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2008-2028 1/	63	42	43	43	44	43	43	41	40	39	**39**	34	26
A2. New public sector loans on less favorable terms in 2008-2028 2	63	44	41	38	36	33	30	27	25	23	**22**	21	21
A3. Conditional delivery of HIPC and MDRI	63	44	41	38	26	25	22	21	19	18	**18**	19	19
A4. Conditional delivery of HIPC, MDRI and beyond HIPC debt relief	63	44	41	38	16	17	17	16	16	16	**16**	18	18
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	63	45	43	40	37	34	30	27	24	22	**21**	20	19
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	63	39	33	31	29	26	23	20	18	16	**15**	16	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	63	42	43	40	37	34	30	27	24	22	**21**	20	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	63	48	48	45	42	38	35	31	29	26	**25**	22	20
B5. Combination of B1-B4 using one-half standard deviation shocks	63	37	34	31	29	26	23	21	18	16	**16**	16	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	63	66	61	57	53	49	43	39	35	32	**30**	29	27
PV of debt-to-exports ratio													
Baseline	135	116	109	101	95	87	74	64	55	49	**45**	39	38
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2008-2028 1/	135	109	113	115	117	115	108	100	94	88	**84**	68	54
A2. New public sector loans on less favorable terms in 2008-2028 2	135	115	109	102	95	88	75	65	57	51	**47**	42	42
A3. Conditional delivery of HIPC and MDRI	135	116	109	101	70	65	57	50	44	40	**38**	38	38
A4. Conditional delivery of HIPC, MDRI and beyond HIPC debt relief	135	116	109	101	42	44	42	39	38	35	**34**	36	37
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	135	116	108	101	94	86	73	63	54	48	**44**	39	37
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	135	82	77	71	67	60	50	42	36	31	**29**	28	29
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	135	116	108	101	94	86	73	63	54	48	**44**	39	37
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	135	125	126	118	111	102	88	76	66	59	**54**	45	41
B5. Combination of B1-B4 using one-half standard deviation shocks	135	79	70	65	60	55	46	39	33	29	**26**	25	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	135	116	108	101	94	86	73	63	54	48	**44**	39	37
PV of debt-to-revenue ratio													
Baseline	333	234	260	204	216	197	175	154	134	118	**111**	97	86
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2008-2028 1/	333	219	270	231	265	260	253	240	227	213	**207**	166	122
A2. New public sector loans on less favorable terms in 2008-2028 2	333	233	261	205	215	197	177	156	138	123	**116**	102	95
A3. Conditional delivery of HIPC and MDRI	333	234	260	204	158	148	133	119	106	96	**94**	92	87
A4. Conditional delivery of HIPC, MDRI and beyond HIPC debt relief	333	234	260	204	96	100	98	95	91	85	**85**	87	84
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	333	238	270	211	223	203	180	157	138	121	**113**	99	88
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	333	208	211	164	172	155	135	116	100	86	**81**	79	76
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	333	223	270	211	223	203	180	157	137	121	**113**	99	88
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	333	252	302	238	251	230	206	182	161	143	**134**	109	92
B5. Combination of B1-B4 using one-half standard deviation shocks	333	198	214	166	175	158	138	119	102	89	**83**	79	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	333	351	388	304	320	291	258	226	197	173	**163**	142	126

Table 1b.Cote d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028 (continued)
(In percent)

	Projections												
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	**2018**	2023	2028
Debt service-to-exports ratio													
Baseline	4	6	8	8	12	13	11	10	9	6	**4**	2	2
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2008-2028 1/	4	6	8	7	11	13	12	11	10	7	**5**	4	4
A2. New public sector loans on less favorable terms in 2008-2028 2	4	7	8	8	12	13	12	11	10	7	**4**	3	2
A3. Conditional delivery of HIPC and MDRI	4	6	8	8	9	10	9	8	7	5	2	2	2
A4. Conditional delivery of HIPC, MDRI and beyond HIPC debt relief	4	6	8	8	5	5	5	5	4	3	2	1	2
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	4	7	8	8	12	13	11	10	9	6	**4**	2	2
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	4	6	7	6	10	11	10	9	8	5	**3**	1	1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	4	7	8	8	12	13	11	10	9	6	**4**	2	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	4	7	9	8	12	13	11	10	9	7	**4**	3	2
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	6	6	9	10	8	8	7	5	**3**	1	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	4	7	8	8	12	13	11	10	9	6	**4**	2	2
Debt service-to-revenue ratio													
Baseline	9	15	20	16	26	29	26	24	22	15	**9**	6	5
A. Alternative Scenarios													
A1. Key variables at their historical averages in 2008-2028 1/	9	13	18	15	26	29	28	26	25	18	**12**	10	9
A2. New public sector loans on less favorable terms in 2008-2028 2	9	15	20	16	27	30	28	26	24	17	**11**	7	5
A3. Conditional delivery of HIPC and MDRI	9	15	20	16	20	24	21	20	18	11	5	4	5
A4. Conditional delivery of HIPC, MDRI and beyond HIPC debt relief	9	15	20	16	11	12	11	11	10	8	5	3	5
B. Bound Tests													
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	9	15	21	16	27	30	28	26	23	16	**10**	6	5
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	9	15	20	15	26	28	26	24	22	15	**9**	4	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	9	14	21	16	27	30	28	25	23	16	**10**	6	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	9	15	21	16	27	30	27	25	23	16	**10**	7	5
B5. Combination of B1-B4 using one-half standard deviation shocks	9	13	19	15	25	28	25	24	22	15	**9**	4	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	9	22	30	23	39	44	40	37	34	23	**14**	8	7
Memorandum item:													
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	50	50	50	50	50	50	50	50	50	50	**50**	50	50

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a.Cote d'Ivoire: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-2028
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2005	2006	2007	Average	Standard Deviation	2008	2009	2010	2011	2012	2013	2008-13 Average	2018	2028	2014-28 Average
Public sector debt 1/	12.8	12.8	78.6			76.3	50.9	46.9	43.8	43.4	42.1		37.7	41.7	
o/w foreign-currency denominated	0.0	0.0	67.8			65.2	40.6	38.0	36.1	36.5	36.0		30.1	32.1	
Change in public sector debt	0.3	0.0	65.7			-2.3	-25.3	-4.0	-3.1	-0.4	-1.3		-0.2	-0.1	
Identified debt-creating flows	...	...	...			-2.0	-1.0	0.1	-2.0	0.2	0.8		1.8	0.5	
Primary deficit	0.2	0.5	-0.3	0.4	1.3	-0.1	-0.7	1.6	-0.2	1.8	2.8	0.9	3.9	2.9	3.6
Revenue and grants	18.1	19.0	19.7			20.6	19.1	16.7	18.9	17.5	17.5		19.8	22.7	
of which: grants	1.1	0.6	0.6			1.7	0.2	0.9	0.2	0.8	0.8		1.0	1.0	
Primary (noninterest) expenditure	18.4	19.5	19.4			20.4	18.4	18.3	18.7	19.2	20.4		23.7	25.5	
Automatic debt dynamics	...	...	...			-1.8	-0.2	-1.5	-1.7	-1.6	-2.0		-2.2	-2.3	
Contribution from interest rate/growth differential	...	...	...			-6.8	-1.8	-1.4	-1.6	-1.5	-1.9		-2.2	-2.4	
of which: contribution from average real interest rate	...	...	...			-5.0	0.9	0.7	0.5	0.7	0.5		-0.2	-0.5	
of which: contribution from real GDP growth	-0.2	-0.1	-0.2			-1.8	-2.7	-2.1	-2.1	-2.2	-2.4		-1.9	-2.0	
Contribution from real exchange rate depreciation	...	...	...			5.0	1.5	-0.1	-0.1	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	...	...	...			-0.4	-24.4	-4.1	-1.2	-0.5	-2.1		-2.0	-0.7	
Other Sustainability Indicators															
PV of public sector debt	12.8	12.8	79.9			79.3	54.6	50.0	45.9	42.7	38.9		28.5	28.1	
o/w foreign-currency denominated	0.0	0.0	69.1			68.3	44.3	41.1	38.1	35.8	32.8		20.9	18.6	
o/w external	...	...	69.1			68.3	44.3	41.1	38.1	35.8	32.8		20.9	18.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	2.3	2.7	5.5			1.6	3.0	5.9	3.6	6.8	8.3		6.1	4.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	70.8	67.5	405.3			385.8	285.4	299.6	242.6	244.9	221.9		144.1	123.9	
PV of public sector debt-to-revenue ratio (in percent)	75.3	69.8	417.3			420.8	288.3	315.8	244.8	256.7	232.9		152.1	129.9	
o/w external 3/	…	…	360.9			362.4	234.0	259.4	203.6	215.2	196.4		111.3	85.8	
Debt service-to-revenue and grants ratio (in percent) 4/	11.6	11.5	29.3			8.5	19.6	25.8	20.5	28.5	31.2		10.9	6.8	
Debt service-to-revenue ratio (in percent) 4/	12.3	11.9	30.1			32.7	28.5	26.6	24.6	17.6	11.5		8.2		
Primary deficit that stabilizes the debt-to-GDP ratio	-0.1	0.5	-66.1			2.2	24.6	5.6	2.9	2.2	4.2		4.1	3.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.9	0.7	1.6	0.4	2.5	2.3	3.7	4.2	4.7	5.3	5.8	4.3	5.4	4.9	5.1
Average nominal interest rate on forex debt (in percent)	...	...	...	474.0	553.8	0.4	1.4	4.0	4.0	4.7	4.0	3.1	1.9	1.0	1.8
Average real interest rate on domestic debt (in percent)	...	...	...	…	…	-2.6	3.8	1.7	1.9	2.1	2.3	1.5	2.7	2.9	2.9
Real exchange rate depreciation (in percent, + indicates depreciation)	13.8	-10.2	-9.5	-4.0	11.5	8.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.6	4.5	2.8	2.9	1.9	8.0	0.5	2.7	2.9	2.9	2.9	3.3	3.4	3.1	3.2
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.0	0.0	0.1	0.1	-0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	45.0	31.5	49.1	51.1	44.8	46.8	44.7	55.0	57.3	..

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Cote d'Ivoire: Sensitivity Analysis for Key Indicators of Public Debt 2008-2028

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
PV of Debt-to-GDP Ratio								
Baseline	79	55	50	46	43	39	28	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	79	57	54	52	49	46	28	15
A2. Primary balance is unchanged from 2008	79	55	49	45	41	35	15	3
A3. Permanently lower GDP growth 1/	79	55	51	47	44	41	33	42
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	79	59	58	55	53	49	42	50
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	79	56	52	47	44	40	29	29
B3. Combination of B1-B2 using one half standard deviation shocks	79	58	56	52	50	46	39	45
B4. One-time 30 percent real depreciation in 2009	79	85	78	72	66	58	38	29
B5. 10 percent of GDP increase in other debt-creating flows in 2009	79	61	56	51	48	44	32	30
PV of Debt-to-Revenue Ratio 2/								
Baseline	386	285	300	243	245	222	144	124
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	386	298	320	273	281	257	135	61
A2. Primary balance is unchanged from 2008	386	287	295	239	235	202	77	0
A3. Permanently lower GDP growth 1/	386	287	304	248	253	232	165	183
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	386	306	347	291	300	279	212	220
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	386	293	309	250	253	229	149	126
B3. Combination of B1-B2 using one half standard deviation shocks	386	304	332	276	283	263	195	197
B4. One-time 30 percent real depreciation in 2009	386	444	469	380	375	333	190	128
B5. 10 percent of GDP increase in other debt-creating flows in 2009	386	317	334	272	274	249	162	133
Debt Service-to-Revenue Ratio 2/								
Baseline	8	20	26	20	28	31	11	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	20	28	23	33	38	11	0
A2. Primary balance is unchanged from 2008	8	20	26	20	28	31	6	0
A3. Permanently lower GDP growth 1/	8	20	26	21	29	32	12	11
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	8	21	29	24	33	37	15	13
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	8	20	26	21	29	32	11	7
B3. Combination of B1-B2 using one half standard deviation shocks	8	21	28	23	32	35	14	12
B4. One-time 30 percent real depreciation in 2009	8	23	34	28	40	45	17	11
B5. 10 percent of GDP increase in other debt-creating flows in 2009	8	20	27	23	31	34	11	8

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the length of the projection period.
2/ Revenues are defined inclusive of grants.